

AP-E18 Graph-Tight Annuli on a Minimizing Diagonal: Failure of the Reverse-Hölder Import, a Boundary-Straddling Rank-Two Needle, and the Scale-Polished Repair

Route-F derivation ledger

24 July 2026

Abstract

AP-E17 reduced the volumetric relaxation gate to the graph-tight annulus condition (GTA). This note tests whether GTA follows from the actual minimizing diagonal, rather than from an arbitrary bounded graph sequence. The answer separates two notions of minimality.

The AP-E17 diagonal has an *additive* comparison error bounded by its global energy excess. After normalization on a tangent ball this error is $o(1)$, but it is not controlled relative to every unresolved microscopic subball. Consequently it gives no uniform reverse-Hölder seed. Standard (p, q) -growth regularity cannot be imported: in dimension three the current ambient bounds have $(p, q) = (2, 6)$, outside the 2024 relaxed-quasiconvex range $q < \min\{3, 3\} = 3$. Treating the complete-minor vector as an $\mathcal{A}$ -free quadratic field also fails, because the admissible graph is cut out by nonlinear Plücker identities and the $\mathbb{S}^3$ constraint; the map is not minimizing over the full linear $\mathcal{A}$ -free class.

More decisively, we construct a smooth, boundary-straddling rank-two needle. In tangent variables its primary amplitude, secondary amplitude, and radius are

$$A_\varepsilon = \varepsilon^{-1/2}, \quad b_\varepsilon = h_\varepsilon = \varepsilon^2.$$

It has strong L^2 mass $O(\varepsilon^5)$, first- and second-minor energy $O(\varepsilon)$, and zero pure third minor. Nevertheless

$$\int |z_\varepsilon|^2 |Dz_\varepsilon|^2 \asymp 1, \quad \int |z_\varepsilon|^2 |M_2(Dz_\varepsilon)|^2 \asymp 1.$$

Placed across the boundary of the tangent ball, it defeats *every* shrinking annulus thickness. At physical tangent radius $r_\varepsilon = \varepsilon$, its target amplitude is only $r_\varepsilon A_\varepsilon = \sqrt{\varepsilon}$ and its added energy is $O(r_\varepsilon^4) = o(r_\varepsilon^3)$. It is compactly homotopic to the identity, preserves degree, does not change the tangent defect measure, and remains within the normalized additive quasiminimality error.

Thus GTA is false as a universal consequence of the AP-E17 *additive* minimizing-diagonal axioms. This does not disprove the existence of a specially selected GTA recovery sequence or regularity of an exact local minimizer. The correct replacement gate is now explicit: construct a scale-polished recovery sequence with a comparison gauge that is uniform down to the needle scale, or prove the corresponding energy-amplitude tightness directly. All downstream promotions remain closed.

Contents

1	The exact logical input from AP-E17	2
2	Why a standard reverse-Hölder theorem cannot be imported	2
2.1	The ambient (p, q) exponent test	2
2.2	The tempting $\mathcal{A}$ -free lift and its obstruction	2
3	The boundary-straddling rank-two needle	3
4	Failure for every shrinking annulus	3
5	Compatibility with an additive minimizing diagonal	4
6	The corresponding reverse-Hölder no-go	4
7	The scale-polished repair	4
8	Gate decision	5

1 The exact logical input from AP-E17

The frozen complete-minor density is

$$W(F) = \frac{1}{2} (|F|^2 + R|M_2(F)|^2 + K|M_3(F)|^2), \quad R = 1, \quad K = 0.35. \quad (1)$$

Let u_{j_ℓ} be the AP-E17 diagonal at a volumetric tangent point x_0 , with tangent radii $r_\ell \downarrow 0$. Its local relative-homotopy deficit satisfies, for every fixed $0 < \rho < 1$,

$$0 \leq \epsilon_{j_\ell}(B_{\rho r_\ell}(x_0)) \leq \mathcal{E}(u_{j_\ell}) - I_B = o(\mu(B_{r_\ell}(x_0))) = o(r_\ell^3). \quad (2)$$

This is a strong statement at the tangent-ball scale. It is only an *additive* statement: if $s_\ell \ll r_\ell$, then (2) gives no useful estimate for $\epsilon_{j_\ell}(B_{s_\ell})/s_\ell^3$.

In scaled coordinates put

$$v_\ell(y) = \frac{u_{j_\ell}(x_0 + r_\ell y) - q}{r_\ell}, \quad v_0(y) = F_0 y, \quad z_\ell = v_\ell - v_0. \quad (3)$$

The GTA weighted term is

$$\mathcal{A}_\ell(\delta) = \frac{1}{\delta^2} \int_{A_{2\delta}} |z_\ell|^2 (1 + R|Dv_\ell|^2 + K|M_2(Dv_\ell)|^2) \, dy. \quad (4)$$

AP-E17 proves the full boundary transfer if there exists $\delta_\ell \downarrow 0$ for which (4) tends to zero, the annular graph energy vanishes, and the annulus stays in one target hemisphere.

2 Why a standard reverse-Hölder theorem cannot be imported

2.1 The ambient (p, q) exponent test

As a function of the ordinary derivative,

$$c|F|^2 \leq 1 + W(F) \leq C(1 + |F|^6). \quad (5)$$

Thus the coarse ambient exponents are $(p, q) = (2, 6)$. The 2024 partial regularity theorem for relaxed strongly quasiconvex (p, q) functionals requires

$$1 \leq p \leq q < \min \left\{ \frac{np}{n-1}, p+1 \right\}. \quad (6)$$

For $n = 3$ and $p = 2$ the right side is 3, whereas $q = 6$. Moreover, that theorem explicitly does not derive the higher-gradient integrability which would be needed to bridge this gap [1].

2.2 The tempting $\mathcal{A}$ -free lift and its obstruction

The complete-minor vector

$$\Xi(Du) = (Du, \sqrt{R} M_2(Du), \sqrt{K} M_3(Du)) \quad (7)$$

does satisfy linear differential identities: the first block is curl free, and the Hodge duals of the second-minor forms are divergence free. Since $W = |\Xi|^2/2$, the 2025 $\mathcal{A}$ -free higher-integrability theorem is a natural candidate [2].

It does not apply to the present minimizer. The range of Ξ is not the full kernel of a linear constant-rank operator. It also satisfies

$$M_2 = Du \wedge Du, \quad M_3 = Du \wedge Du \wedge Du, \quad |u| = 1, \quad u^T Du = 0, \quad (8)$$

which are nonlinear algebraic and target constraints. Local minimality is known only among fields obeying (8), not among all $\mathcal{A}$ -free variations. Enlarging the competitor class would assume the very graph-recovery theorem under investigation.

Proposition 2.1 (Reverse-Hölder import verdict). *Neither (6) nor the quadratic $\mathcal{A}$ -free theorem supplies a reverse-Hölder estimate for the AP-E17 minimizing diagonal. Any valid proof must use new scale-aware minimality on the nonlinear complete-minor graph.*

3 The boundary-straddling rank-two needle

Fix a point $y_* \in \partial B_1$. In a boundary chart, choose $\psi, \phi \in C_c^\infty(B_1)$ such that on a smaller open set crossing the flat interface through the origin,

$$|\psi| + |D\psi| \geq c, \quad |D\psi \wedge D\phi| \geq c. \quad (9)$$

Take two orthonormal target directions $e_1, e_2 \in T_q \mathbb{S}^3$. For $\varepsilon \downarrow 0$ set

$$A_\varepsilon = \varepsilon^{-1/2}, \quad b_\varepsilon = \varepsilon^2, \quad h_\varepsilon = \varepsilon^2, \quad (10)$$

and define the tangent perturbation

$$z_\varepsilon(y) = A_\varepsilon \psi \left(\frac{y - y_*}{h_\varepsilon} \right) e_1 + b_\varepsilon \phi \left(\frac{y - y_*}{h_\varepsilon} \right) e_2. \quad (11)$$

The support is a ball of radius h_ε straddling ∂B_1 . On the set in (9), Dz_ε has rank exactly two.

Theorem 3.1 (Exact needle scaling). *Up to profile-dependent positive constants,*

$$\int |z_\varepsilon|^2 \asymp A_\varepsilon^2 h_\varepsilon^3 = \varepsilon^5, \quad (12)$$

$$\int |Dz_\varepsilon|^2 \asymp A_\varepsilon^2 h_\varepsilon = \varepsilon, \quad (13)$$

$$\int |M_2(Dz_\varepsilon)|^2 \asymp \frac{A_\varepsilon^2 b_\varepsilon^2}{h_\varepsilon} = \varepsilon, \quad (14)$$

$$M_3(Dz_\varepsilon) = 0. \quad (15)$$

Nevertheless,

$$\int |z_\varepsilon|^2 |Dz_\varepsilon|^2 \asymp A_\varepsilon^4 h_\varepsilon = 1, \quad (16)$$

$$\int |z_\varepsilon|^2 |M_2(Dz_\varepsilon)|^2 \asymp A_\varepsilon^2 \frac{A_\varepsilon^2 b_\varepsilon^2}{h_\varepsilon} = 1. \quad (17)$$

Proof. The support volume is comparable to h_ε^3 . The two derivative sizes are $A_\varepsilon/h_\varepsilon$ and $b_\varepsilon/h_\varepsilon = 1$. Equations (12)–(14) follow by multiplying the corresponding pointwise powers by the support volume. The derivative has only two target rows, so every pure third minor is zero. On the active profile set, $|z_\varepsilon| \asymp A_\varepsilon$; multiplying (13) and (14) by A_ε^2 gives (16)–(17). $\square$

Adding a bounded affine matrix F_0 does not alter these powers. Since the rank-two update has no term cubic in itself,

$$\begin{aligned} M_2(F_0 + Dz) - M_2(F_0) &= L_2(F_0, Dz) + M_2(Dz), \\ M_3(F_0 + Dz) - M_3(F_0) &= L_3(F_0, F_0, Dz) + L_3(F_0, Dz, Dz), \end{aligned} \quad (18)$$

and every displayed contribution has squared integral $O(\varepsilon)$.

4 Failure for every shrinking annulus

Let $0 < \delta_\varepsilon < 1/4$. The active region in the intersection of the needle and $A_{2\delta_\varepsilon}$ has volume bounded below by

$$ch_\varepsilon^3 \quad \text{if } \delta_\varepsilon \geq h_\varepsilon, \quad ch_\varepsilon^2 \delta_\varepsilon \quad \text{if } \delta_\varepsilon < h_\varepsilon.$$

It follows from (16) that

$$\mathcal{A}_\varepsilon(\delta_\varepsilon) \geq \begin{cases} c \delta_\varepsilon^{-2}, & \delta_\varepsilon \geq h_\varepsilon, \\ c(\varepsilon^2 \delta_\varepsilon)^{-1}, & 0 < \delta_\varepsilon < h_\varepsilon. \end{cases} \quad (19)$$

Both branches diverge for every choice $\delta_\varepsilon \downarrow 0$. In contrast, the unweighted annular graph energy obeys

$$\int_{A_{2\delta_\varepsilon}} W(F_0 + Dz_\varepsilon) \leq C \begin{cases} \varepsilon, & \delta_\varepsilon \geq h_\varepsilon, \\ \delta_\varepsilon/\varepsilon, & \delta_\varepsilon < h_\varepsilon, \end{cases} \longrightarrow 0, \quad (20)$$

because the second branch has $\delta_\varepsilon < h_\varepsilon = \varepsilon^2$.

Theorem 4.1 (GTA instability). *The strong tangent convergence, vanishing annular graph energy, and vanishing normalized added energy do not imply GTA, even for a genuinely rank-two perturbation whose physical image stays in one target hemisphere.*

5 Compatibility with an additive minimizing diagonal

Take the physical tangent radius to be

$$r_\varepsilon = \varepsilon. \quad (21)$$

First insert (11) through the exponential chart over the bounded affine background $v_0 = F_0 y$. More generally the same calculation holds on a recovery patch whose background complete-minor graph is equiintegrable at the needle scale. Its largest target displacement is

$$r_\varepsilon A_\varepsilon = \sqrt{\varepsilon} \rightarrow 0, \quad (22)$$

so radial projection and its exterior powers have uniform constants. The physical added derivative energy is

$$r_\varepsilon^3 O(\varepsilon) = O(r_\varepsilon^4) = o(r_\varepsilon^3). \quad (23)$$

The perturbation is supported in a contractible target chart and the homotopy obtained by multiplying (11) by $t \in [0, 1]$ is compact. It therefore preserves the exterior trace and global degree.

Theorem 5.1 (Instability of the AP-E17 additive tangent axioms). *On the affine tangent background, the rank-two needle has $o(r_\varepsilon^3)$ physical cost. It does not change the weak tangent graph measure, and its removable energy is compatible with*

$$\frac{\epsilon_{j\ell}(B_{\rho r_\ell})}{\mu(B_{r_\ell})} \rightarrow 0 \quad (24)$$

for every fixed ρ whenever it is inserted into a controlled recovery patch. Nevertheless it violates (4) for every $\delta_\ell \downarrow 0$.

Proof. For the affine background, (18) and (13)–(14) make the normalized graph-energy change $O(\varepsilon)$; hence it disappears from the weak tangent measure and is $o(1)$ at the additive comparison normalization. The same conclusion holds on a patch for which the background mixed terms in (18) are equiintegrable at that scale. Degree and trace are unchanged by the compact homotopy. Equation (19) proves failure of the remaining GTA condition. $\square$

This theorem has a deliberately precise scope. It disproves a theorem which tries to derive GTA from the AP-E17 additive tangent estimates alone: those estimates cannot distinguish the needle. It does not prove that the repository's selected diagonal actually contains the needle, that no carefully selected recovery sequence has GTA, or that the degree-one minimizer has a nonzero relaxation defect.

6 The corresponding reverse-Hölder no-go

On its active set the needle graph density satisfies

$$G_\varepsilon \asymp \varepsilon^{-5}, \quad |\text{supp } G_\varepsilon| \asymp \varepsilon^6, \quad \int G_\varepsilon \asymp \varepsilon. \quad (25)$$

For every $\sigma > 0$,

$$\int G_\varepsilon^{1+\sigma} \asymp \varepsilon^{1-5\sigma}, \quad \frac{(\int G_\varepsilon^{1+\sigma})^{1/(1+\sigma)}}{\int G_\varepsilon} \asymp \varepsilon^{-6\sigma/(1+\sigma)} \rightarrow \infty. \quad (26)$$

Thus no sequence-uniform fixed-ball $L^{1+\sigma}$ gain follows from (2). At the needle scale the removable energy is $O(\varepsilon)$, exactly the size allowed by the additive error.

7 The scale-polished repair

Definition 7.1 (Uniform microscale comparison gauge). *A tangent recovery sequence is scale polished if there are $\omega_\ell \downarrow 0$ and $s_\ell \downarrow 0$ such that for every ball $B_s(y) \subseteq B_1$ with $s_\ell \leq s \leq 1/4$,*

$$\epsilon_\ell(B_s(y)) \leq \omega_\ell s^3. \quad (27)$$

An exact local minimizer has $\omega_\ell = 0$.

The needle violates (27) at $s = h_\varepsilon = \varepsilon^2$, since its removable energy is $\asymp \varepsilon$ whereas $s^3 = \varepsilon^6$. Thus (27) detects exactly what the global additive estimate misses. It is not yet proved that a degree-preserving recovery sequence satisfying (27) exists.

A second, weaker sufficient target is the energy–amplitude tightness

$$\int_{B_1} |z_\ell|^2 (1 + R|Dv_\ell|^2 + K|M_2(Dv_\ell)|^2) dy \longrightarrow 0. \quad (28)$$

Together with no boundary mass and a one-hemisphere collar, (28) permits a slow diagonal choice $\delta_\ell \downarrow 0$ with

$$\delta_\ell^{-2} \times \text{the left side of (28)} \rightarrow 0$$

and vanishing annular graph energy, hence GTA. The needle has left side $\asymp 1$ and is the sharp obstruction.

8 Gate decision

The requested GTA investigation therefore closes with a negative universal result and a narrower positive target:

1. **Reverse-Hölder from the current diagonal: false.** The additive error is not scale uniform; current (p, q) and $\mathcal{A}$ -free theorems do not apply.
2. **GTA from the additive tangent axioms alone: false.** The boundary-straddling rank-two needle is invisible to those normalized limits while violating every annulus thickness. Whether the selected repository diagonal contains or can be polished to exclude it remains open.
3. **GTA for a scale-polished diagonal: open.** The next theorem must construct (27), or prove (28) plus the hemisphere collar by another exact local-selection argument.

The counterexample does not reopen lattice work and does not authorize the bosonic Hessian. Full $\mu = 0$, local recovery, classicality, continuum isolation, determinant promotion, and the degree-one portal remain false. Parallel Dirac/Callias mathematics on a prescribed smooth background remains allowed.

The independent verifier

`route_f/code/verify_ap_e18_gta_minimizing_diagonal.py`

checks the rank, every scaling exponent, both annulus branches, reverse-Hölder divergence, literature applicability, and all fail-closed gates.

References

- [1] F. Gmeineder and J. Kristensen, Quasiconvex functionals of (p, q) -growth and the partial regularity of relaxed minimizers, *Arch. Rational Mech. Anal.* **248** (2024), 80, <https://doi.org/10.1007/s00205-024-02013-8>.
- [2] S. Schiffer, $\mathcal{A}$ -free truncation and higher integrability of minimisers, *Calc. Var. Partial Differential Equations* **64** (2025), 30, <https://doi.org/10.1007/s00526-024-02855-w>.